

SUPPLEMENTARY

Elementary Number Theory

Instructor: Ramdin Mawia

Marks: 30

Time: October 8, 2024; 10:00–12:00.

Attempt all the THREE problems. The maximum you can score is 18.

1. Do there exist integers a and b such that $11a^2 + 169b = 109$? Give a complete justification with examples, if any. **10**
2. Prove that if a, b and c are integers such that $a|bc$, then $a|\gcd(a, b)\gcd(a, c)$. **10**
3. Describe all primes $p > 5$ for which -5 is a quadratic residue mod p . **10**